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SORTING SYSTEM

Abstract

A sorting system includes: a sorting platform, a first tag reader, a control server, at least one sorting robot, at least one feeding table and a plurality of collection containers. The sorting platform includes one or more layers; each feeding table is configured to place at least one item to be sorted; the first tag reader is configured to read and send tag information in a tag of each item to be sorted to the control server; the control server is configured to determine a sorting destination of the item to be sorted, determine a target collection container corresponding to the item to be sorted, and send a delivery instruction and a delivery path of the item to be sorted; and the sorting robot is configured to, in response to the delivery instruction, deliver the item to be sorted to the target collection container according to the delivery path.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The application is a continuation of U.S. patent application Ser. No. 17/636,660, filed on Jul. 26, 2022, which is a national stage entry under 35 U.S.C. § 371 of International Application No. PCT/CN2020/081270, filed on Mar. 26, 2020, which claims priority to Chinese Patent Application No. 201910765281.2, filed on Aug. 19, 2019, the entire disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a field of logistics and warehousing technologies, and particularly to a sorting system and method.

BACKGROUND

[0003] In the warehousing and logistics industry, sorting of commodities or packages is a very important link in the warehousing and logistics industry. Through the sorting, commodities or packages with different destinations can be collected and distributed, so as to achieve unified collection, storage and transportation of commodities or packages of the same destination.

[0004] At present, in the sorting process, a manual hand-held barcode identification device is pointed at the barcode of the commodity or package to identify the barcode of the commodity or package. However, with the sharp increase in the number and types of the commodities or packages, the task of the sorting operation is getting heavier and heavier, and the proportion of the sorting operation in the warehousing and logistics industry is increasing. If the traditional method is used, it will not only greatly increase the labor cost, but also cause incorrect identification problems due to such as barcode wrinkles, which will seriously affect the sorting efficiency.

SUMMARY

[0005] Embodiments of the present disclosure provide a sorting system, including: a sorting platform, a first tag reader, a control server, at least one sorting robot, at least one feeding table and a plurality of collection containers.

[0006] The sorting platform includes one or more layers, and the at least one sorting robot runs on the sorting platform; [0007] each feeding table is configured to place at least one item to be sorted; and each item to be sorted is provided with a tag including sorting destination information of the item to be sorted; [0008] the first tag reader is configured to read tag information in the tag, and send the tag information to the control server; [0009] the plurality of collection containers are divided into at least one collection container group; [0010] the control server is configured to determine a sorting destination of the item to be sorted according to the tag information including the sorting destination information, determine a target collection container corresponding to the item to be sorted from the collection container group according to the sorting destination, and send a delivery instruction and a delivery path of the item to be sorted planned according to a position of the target collection container to the sorting robot; and [0011] the sorting robot is configured to, in response to the delivery instruction received from the control server, deliver the item to be sorted to the target collection container according to the delivery path. Embodiments of the present disclosure further provide a sorting system, including: a first tag reader, a control server, at least one sorting robot, at least one feeding table, a plurality of collection container groups, a first grabbing component and a sorting platform.

[0012] Each feeding table is configured to place at least one item to be sorted; and each item to be sorted is provided with a tag including sorting destination information of the item to be sorted; [0013] the first tag reader is configured to read tag information in the tag, and send the tag information to the control server to determine a sorting destination of the item to be sorted; [0014] the first grabbing component is configured to receive the at least one item to be sorted, and to place the at least one item to be sorted on the at least one sorting robot; and [0015] the at least one sorting robot is configured to, in response to a delivery instruction received from the control server, deliver the at least one item to be sorted to a target collection container in a collection container group according to a delivery path. Embodiments of the present disclosure further provide a sorting system, including: a first tag reader, a control server, at least one sorting robot, at least one feeding table, and a plurality of collection containers.

[0016] Each feeding table is configured to place at least one item to be sorted; and each item to be sorted is provided with a tag including sorting destination information of the item to be sorted; [0017] the first tag reader is configured to read tag information in the tag, and send the tag information to the control server; [0018] the control server is configured to determine a sorting destination of the item to be sorted according to the tag information including the sorting destination information, determine a target collection container corresponding to the item to be sorted according to the sorting destination, and send a delivery instruction and a delivery path of the item to be sorted planned according to a position of the target collection container to the sorting robot; and [0019] the sorting robot is configured to, in response to the delivery instruction received from the control server, deliver the item to be sorted to the target collection container according to the delivery path.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The drawings are for the purpose of illustrating alternative embodiments only and are not to be considered as limitation to the present disclosure. Also, the same components are denoted by the same reference numerals throughout the drawings. In the attached drawings:

[0021] FIG. 1 is a schematic diagram of a sorting system provided in an embodiment of the present disclosure.

[0022] FIG. 2 is a schematic diagram of a floor-type sorting scenario of a sorting system provided in an embodiment of the present disclosure.

[0023] FIG. 3 is a schematic diagram of another sorting system provided in an embodiment of the present disclosure.

[0024] FIG. 4 is a schematic diagram of another sorting system provided in an embodiment of the present disclosure.

[0025] FIG. 5 is a schematic diagram showing a part of an item transfer facility in a sorting system provided in an embodiment of the present disclosure.

[0026] FIG. 6 is a schematic diagram showing a part of floor-type sorting scenario including a collection table provided in an embodiment of the present disclosure.

[0027] FIG. 7 is a schematic diagram of another sorting system provided in an embodiment of the present disclosure.

[0028] FIG. 8 is a flowchart of a sorting method provided in an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0029] The present application will be described below with reference to the accompanying drawings and embodiments. It should be understood that the specific embodiments described herein are only used to explain the present application, but not to limit the present application. In

addition, it should be noted that, for the convenience of description, the drawings only show some but not all the structures related to the present application.

[0030] The following briefly introduces the overall sorting process of items to be sorted (for example, commodities or packages to be sorted) in embodiments. The overall sorting process includes three links: a feeding link, a delivery link, and a collection link. In embodiments, the feeding link realizes that the items to be sorted are distributed from the transport equipment to feeding tables; the delivery link realizes that the items to be sorted on the feeding tables are delivered to collection containers deployed in the delivery area; the collection link realizes transfer of the collection containers containing the items to be sorted to collection tables, and the sorting process of the items to be sorted can be completed after the above three links.

[0031] A sorting system and method provided in embodiments of the present disclosure are described below through the following embodiments and their optional solutions.

[0032] FIG. 1 is a schematic diagram of a sorting system provided in an embodiment of the present disclosure. The technical solution of this embodiment can be applied to a case of sorting commodities or packages in a sorting scenario, for example, in a case of realizing sorting of commodities or packages in an unmanned sorting scenario. The sorting system can implement the sorting method provided in any embodiment of the present disclosure. As shown in FIG. 1, the sorting system **100** provided in this embodiment of the present disclosure includes at least one radio frequency tag **110**, at least one first radio frequency tag reader **120**, at least one control server **130**, at least one sorting robot **140**, at least one feeding table **160** and at least one collection container **170**.

[0033] At least one feeding table **160** is configured to be placed with at least one item to be sorted **150**. At least one radio frequency tag **110** is configured to be disposed on at least one item to be sorted **150**, and tag information of each radio frequency tag **110** includes sorting destination information of the item to be sorted **150** where each radio frequency tag is located.

[0034] The first radio frequency tag reader **120** is configured to read tag information in the radio frequency tag **110**, and send the read tag information to the control server **130**.

[0035] The control server **130** is configured to determine the sorting destination of the item to be sorted **150** where each radio frequency tag is located according to the tag information read by the first radio frequency tag reader **120**, and determine a target collection container from the at least one collection container **170** according to the sorting destination, and send a delivery instruction to at least one sorting robot **140**.

[0036] The at least one sorting robot **140** is configured to, in response to the delivery instruction, deliver the at least one item to be sorted **150** obtained from the at least one feeding table **160** to the target collection container.

[0037] The working process of the sorting system of this embodiment will be described below with reference to the schematic diagram of the sorting system shown in FIG. 1.

[0038] In this embodiment, the sorting system and method of the present disclosure can be applied in a floor-type sorting scenario or a steel platform sorting scenario, and can deliver the items to be sorted in any of the above-mentioned sorting scenarios. In the floor-type sorting scenario, the sorting robot runs on the ground; and in the steel platform sorting scenario, the sorting robot runs on the steel platform structure.

[0039] In this embodiment, FIG. 2 is a schematic diagram of a floor-type sorting scenario of a sorting system provided in an embodiment of the present disclosure. Referring to FIG. 1 and FIG. 2, a plurality of collection containers **170** are deployed in the middle of the sorting area, and the multiple collection containers **170** are arranged in the sorting area in an array to form a collection container area. For the floor-type sorting scenario, horizontal and vertical gaps between adjacent collection containers **170** constitute walking passage of the sorting robot on the ground. In addition, for the steel platform sorting scenario, each grid opening of the steel platform structure corresponds to a collection container, and horizontal and vertical gaps between adjacent collection containers

can ensure the adjacent grid openings on the steel platform remain a certain horizontal gap and vertical gap, so that the horizontal gaps and vertical gaps between the grid openings on the steel platform constitute the walking passage of the sorting robot on the steel platform. It can be understood that, in the steel platform sorting scenario, the sorting robot needs to run on the steel platform, and the steel platform has a certain height, so the sorting robot generally cannot leave the steel platform, and thus the feeding table can be disposed adjacent to one side of the steel platform, which is convenient for the sorting robot to obtain the items to be sorted from the feeding table. In the floor-type sorting scenario, since the sorting robot is driving on the ground, the above problems do not need to be considered.

[0040] In this embodiment, referring to FIG. 1, the control server **130** refers to a software and/or hardware system with data information storage and data information processing capabilities, and can communicate with the hardware device or software system in the sorting system in a wired or wireless manner. The control server **130** can be used to implement the control and scheduling of the entire sorting system. Optionally, the control server **130** can receive information sent by multiple hardware devices or software systems in the sorting system, and can also send tasks to multiple hardware devices or software systems in the sorting system to count a sorting situation, detect a working state of the sorting system, pass information to the staff, and issue control instructions to multiple types of robots in the sorting system, etc.

[0041] In this embodiment, for each feeding table **160**, one or more items to be sorted can be placed on each feeding table **160**; and for each item to be sorted placed on the feeding table **160**, each item to be sorted **150** carries a uniquely identified radio frequency tag. In one embodiment, the radio frequency tags carried by the items to be sorted **150** are radio frequency identification (RFID) tags. Each radio frequency tag **110** contains preset tag information, and the tag information includes sorting destination information of the item to be sorted. In one embodiment, the sorting destination includes information of a city where a delivery address of the item to be sorted is located, that is, information of the city to be delivered to. For example, the sorting destination information may include identification data for indicating physical locations such as “Tianjin” and “Beijing”.

[0042] In this embodiment, different items to be sorted may need to be sent to different destinations, and at this time, multiple items to be sorted have different sorting destinations. As described above, the sorting destination information of the item to be sorted is carried in the tag information of the radio frequency tag of the item to be sorted. Referring to FIG. 1, one or more first radio frequency tag readers **120** are deployed in the sorting system. When the item to be sorted **150** placed on the feeding table **160** is close to the first radio frequency tag reader **120**, the first radio frequency tag reader **120** can scan the radio frequency tag **110** carried on the item to be sorted **150** and read the tag information in the radio frequency tag **110**. Moreover, the first radio frequency tag reader **120** establishes a communication connection with the control server **130**, and can send the read tag information in the radio frequency tag **110** to the control server **130**.

[0043] In this embodiment, referring to FIG. 1, the control server **130** can receive the tag information of the radio frequency tag **110** sent by the first radio frequency tag reader **120**. Furthermore, the control server **130** can determine the sorting destination of the item to be sorted **150** according to the tag information read by the first radio frequency tag reader **120**. For at least one collection container **170** deployed in the sorting area, each collection container **170** corresponds to at least one sorting destination. In other words, each collection container **170** can have a correspondence relationship with one or more sorting destinations. At this time, the control server **130** can determine the collection container corresponding to the sorting destination according to the above-mentioned correspondence relationship, that is, can determine the target collection container from the at least one collection container **170** according to the sorting destination, that is, determine the target collection container in the sorting area to which the item to be sorted on the feeding table **160** needs to be delivered.

[0044] In this embodiment, referring to FIG. 1 and FIG. 2, for a plurality of collection containers **170** deployed in the sorting area, the collection containers corresponding to the same sorting destination are arranged adjacent to each other, that is, if two collection containers **170** have the same sorting destination, the two collection containers **170** are arranged adjacent to each other. Optionally, for the multiple collection containers **170** deployed in the sorting area, the multiple collection containers **170** may be divided into at least two collection container groups, and the collection containers **170** corresponding to similar sorting destinations may be deployed in the same collection container group. That is, if two collection containers **170** correspond to similar sorting destinations, the two collection containers **170** can be deployed in the same collection container group. In this embodiment, the same sorting destination means that the sorting destinations are the same in physical location, for example, both are “Tianjin”; and the similar sorting destinations means that the two sorting destinations are physically close to each other, that is, the distance between two sorting destinations is less than a preset threshold, for example, one is “Tianjin” and the other is “Beijing”.

[0045] In this embodiment, referring to FIG. 1 and FIG. 2, optionally, the same feeding table **160** is associated with a unique collection container group, and at this time, the multiple sorting destinations corresponding to the same feeding table **160** are the same as the sorting destinations corresponding to the plurality of collection containers **170** in the collection container group described above, which can ensure that the items to be sorted on the same feeding table **160** are delivered to the same collection container group. Optionally, the same collection container group can be associated with one or more feeding tables **160** at the same time, which can ensure that multiple feeding tables **160** deliver items to the collection containers **170** in the same collection container group at the same time, which can increase a delivery efficiency in the collection container group, thereby improving the overall delivery efficiency.

[0046] In this embodiment, referring to FIG. 1, the control server **130** may generate a delivery instruction for items to be sorted according to the determined target collection container, and send the generated delivery instruction to at least one sorting robot **140** located in the sorting area. Furthermore, the sorting robot **140** can respond to the delivery instruction and deliver at least one item to be sorted **150** obtained from the feeding table **160** to the target collection container. In this way, the collection container **170** can temporarily store the items to be sorted in one or more preset sorting destinations, and can ensure that one or more items to be sorted with sorting destinations that have a correspondence relationship with the collection container **170** are delivered into the collection container **170** for storage.

[0047] Optionally, referring to FIG. 1, the sorting area includes a plurality of sorting robots **140** in standby or working state, and the control server **130** may send a delivery instruction for items to be sorted to one target sorting robot of the plurality of sorting robots **140** located in the sorting area. Optionally, when the target sorting robot receives the delivery instruction sent by the control server **130**, it can also receive a delivery path of items to be sorted planned according to the position of the target collection container and sent by the control server **130**, and deliver at least one item to be sorted **150** obtained from the feeding table **160** according to the delivery path.

[0048] On the basis of the above-mentioned embodiment, optionally, referring to FIG. 1, the first radio frequency tag reader **120** is disposed at a preset position of the feeding table **160**; or, the first radio frequency tag reader **120** is disposed at a preset position on the sorting robot **140**; or, the first radio frequency tag reader **120** is arranged between the feeding table **160** and the at least one collection container **170** and a distance between the first radio frequency tag reader **120** and the preset position of the feeding table **160** is less than a preset value. When at least one item to be sorted **150** obtained from at least one feeding table **160** is assigned to at least one sorting robot **140** in the sorting area, the item to be sorted **150** will pass or approach the position where the first radio frequency tag reader **120** is disposed, such that the first radio frequency tag reader **120** reads the tag information of the radio frequency tag.

[0049] The sorting system **100** of this embodiment further includes a first grabbing component **180**. The following describes the settings of the first radio frequency tag reader **120** and the first grabbing component **180** through several optional examples.

[0050] In an optional example, for the case where the first radio frequency tag reader **120** is disposed at the preset position of the feeding table **160**, the first grabbing component **180** is disposed on the feeding table **160**, and the first grabbing component **180** can grab items to be sorted **150** from the feeding table **160**, and place the grabbed items to be sorted **150** on at least one sorting robot **140** in the sorting area. Optionally, the first radio frequency tag reader **120** is disposed on the first grabbing component **180**, and when the first grabbing component **180** grabs the item to be sorted **150** on the feeding table **160**, the first radio frequency tag reader **120** reads the tag information in the radio frequency tag **110** of the item to be sorted **150**.

[0051] In another optional example, for the case where the first radio frequency tag reader **120** is disposed at a preset position on the sorting robot **140**, the first grabbing component **180** is disposed on the body of the sorting robot **140**, and can move with the sorting robot **140**, so that the sorting robot **140** can autonomously grab the items to be sorted **150** on the feeding table **160** and place them in the preset temporary storage place of its own body to temporarily store the grabbed items to be sorted **150**. Optionally, when at least one sorting robot **140** grabs the items to be sorted on the feeding table **160**, the first radio frequency tag reader **120** disposed on the first grabbing component **180** reads the tag information in the radio frequency tags **110** of the items to be sorted **150**.

[0052] In yet another optional example, for the case where the first radio frequency tag reader **120** is disposed between the feeding table **160** and the at least one collection container **170** and the distance between the first radio frequency tag reader **120** and the preset position of the feeding table **160** is less than the preset value, when at least one sorting robot **140** in the sorting area obtains the item to be sorted **150** grabbed from the feeding table **160** through the first grabbing component **180**, the sorting robot **140** will carry the item to be sorted to approach and pass the first radio frequency tag reader **120** disposed between the feeding table **160** and the at least one collection container **170**. At this time, the first radio frequency tag reader **120** reads the tag information in the radio frequency tag **110** of the item to be sorted **150** carried by the sorting robot **140**.

[0053] Optionally, the first grabbing component **180** may include a robotic arm, and the robotic arm is provided with a robotic gripper or a suction cup, through which the item to be sorted **150** located on the feeding table **160** can be grabbed. It can be understood that the setting positions of the first radio frequency tag reader **120** and the first grabbing component **180** shown in FIG. 1 may deviate from the positions described in the text. For details, please refer to the text description.

[0054] On the basis of the above embodiment, optionally, referring to FIG. 1, the sorting robot **140** includes a plurality of delivery mechanisms, and each delivery mechanism can deliver one or more items to be sorted **150** in the same sorting destination. Optionally, through multiple delivery mechanisms provided by itself, the sorting robot **140** can carry the items to be sorted **150** in one or more sorting destinations when delivering the items to be sorted **150**. In an optional example, the multiple items to be sorted **150** carried by the sorting robot **140** may have the same sorting destination. In this case, the sorting robot **140** may deliver the carried multiple items to be sorted to the same collection container at the same time according to the same sorting destination. In another optional example, the multiple items to be sorted **150** carried by the sorting robot **140** may have different sorting destinations. In this case, the sorting robot **140** may sequentially deliver the carried multiple items to be sorted **150** having different sorting destinations to different collection containers **170** for storage according to different sorting routes.

[0055] For the above two optional examples, if the first radio frequency tag reader **120** is disposed on the sorting robot **140**, before the sorting robot **140** delivers each item to be sorted **150** carried by itself, the first radio frequency tag reader **120** on the sorting robot **140** reads the tag information in the radio frequency tag **110** of each item to be sorted **150**, and sends the read tag information to the

control server **130**. Furthermore, for the tag information in the radio frequency tag of each item to be sorted **150** read above, the control server **130** determines the sorting destination of the item to be sorted **150** according to the tag information read by the first radio frequency tag reader **120** and determines a target collection container for the item to be sorted from the at least one collection container **170** according to the sorting destination of the item to be sorted, and sends a delivery instruction for the item to be sorted **150** to at least one sorting robot **140**. Then, the sorting robot **140** responds to the delivery instruction for the item to be sorted, and delivers at least one item to be sorted obtained from at least one feeding table to the corresponding target collection container, thereby realizing automatic delivery of multiple items to be sorted **150**.

[0056] In embodiment of the present application, an object being close to another object indicates that a distance between one object and another object is less than a preset value.

[0057] According to the sorting solution provided in embodiments of the present disclosure, since the first radio frequency tag reader has no requirement on the reading direction, no matter what attitude the items to be sorted are placed in, the first radio frequency tag reader can read the tag information in the radio frequency tags of the items to be sorted without manual operations, ensuring that the tag information can be automatically identified and sent when the items to be sorted are grabbed from the feeding table, so that the control server can automatically determine the sorting destinations of the grabbed items to be sorted, automatically assign the items to be sorted to the collection containers, and send delivery instructions for the items to be sorted to the sorting robots to instruct the sorting robots to deliver the items to be sorted to the corresponding collection containers, and further the sorting robot can automatically deliver the items to be sorted into the collection containers according to the delivery instructions, thereby realizing the unmanned operation of the delivery process, saving time wasted by manually aligning the barcode of the commodity or package to identify the delivery in the delivery process, improving delivery efficiency and reducing labor costs in the delivery process.

[0058] FIG. **3** is a schematic diagram of another sorting system provided in an embodiment of the present disclosure, and the present embodiment is described on the basis of the above-mentioned embodiment. This embodiment may be combined with the optional solutions in one or more of the foregoing embodiments. Referring to FIG. **3**, based on the above-mentioned embodiments, the sorting system further includes a second radio frequency tag reader **190**. The second radio frequency tag reader **190** is close to the location of the items to be sorted that are to be moved to the feeding table **160**. Each feeding table **160** corresponds to at least one sorting destination, and each feeding table **160** can allow placing items to be sorted **150** in at least one sorting destination.

[0059] The second radio frequency tag reader **190** is configured to read tag information in the radio frequency tag of the item to be sorted that is to be moved to the feeding table **160**, and send the read tag information to the control server **130**.

[0060] The control server **130** is further configured to, determine the sorting destination of the item to be sorted that is to be moved to the feeding table **160** according to the tag information read by the second radio frequency tag reader **190**, determine the feeding table **160** corresponding to the item to be sorted that is to be moved to the feeding table from the at least one feeding table **160** according to the sorting destination, so that the item to be sorted that is to be moved to the feeding table is moved to the corresponding feeding table **160**.

[0061] The specific working process of the sorting system of this embodiment will be described in detail below with reference to the schematic diagram of the sorting system shown in FIG. **3**.

[0062] In this embodiment, referring to FIG. **3**, an unloading area is preset in the sorting area corresponding to the sorting system **100**, and a plurality of items to be sorted can be transported to the unloading area by using transport equipment, and unloaded in the unloading area. The items to be sorted in the unloading area may be the items to be sorted that are to be moved to the feeding table, and the items to be sorted in the unloading area need to be moved to the corresponding feeding tables **160**. The item to be sorted that is unloaded from the transport equipment carries a

uniquely identified radio frequency tag, the radio frequency tag includes corresponding tag information, and the tag information includes the sorting destination information of the item to be sorted. The second radio frequency tag reader **190** may be disposed between the items to be sorted in the unloading area that are to be moved to the feeding table and the position of the feeding table **160**. A wired or wireless communication connection can be established between the second radio frequency tag reader **190** and the control server **130**, and the second radio frequency tag reader **190** can read the tag information in the radio frequency tags of the items to be sorted unloaded from the transport equipment, and send the read tag information to the control server **130**.

[0063] In one embodiment, the items to be sorted that are to be moved to the feeding table cannot be placed at will, and one or more item buffer facilities **1100** may be pre-set in the unloading area. The item buffer facility **1100** can receive and buffer the items to be sorted that are transported by the transport equipment and to be moved to the feeding table, and the second radio frequency tag reader **190** can be disposed between the item buffer facility **1100** and the preset position of the feeding table **160**. In one embodiment, the item buffer facility **1100** may be a building facility or a large buffer container, or a large container for buffering items to be sorted that are unloaded from the transport equipment.

[0064] In this embodiment, referring to FIG. 3, the control server **130** can receive the tag information sent from the second radio frequency tag reader **190**, and then can determine the sorting destinations of the items to be sorted that are to be moved to the feeding table according to the tag information sent by the second radio frequency tag reader **190**, preliminarily classify the items to be sorted that are to be moved to the feeding table, according to the sorting destinations, and determine a corresponding feeding table for the classified one or more items to be sorted that are to be moved to the feeding table, that is, from the plurality of feeding tables, the corresponding feeding table **160** to which the items to be sorted need to be moved. Optionally, the same feeding table **160** may be associated with items to be sorted in at least one sorting destination, in other words, the feeding table **160** may temporarily store the items to be sorted in one or more sorting destinations, so that it can be ensured that the items to be sorted with the preset destinations will be initially supplied to the same feeding table **160** for centralized temporary storage, so that batch delivery can be performed from the feeding table in the future.

[0065] In one embodiment, the tag information in the radio frequency tag read by the first radio frequency tag reader **120** is consistent in content with the tag information in the radio frequency tag read by the second radio frequency tag reader **190**, for example, both include the pickup destination of the item to be sorted. The difference between the two is that the functions of the read tag information in the radio frequency tag are different. The tag information read by the first radio frequency tag reader **120** is used for delivery, that is, to determine the collection containers to which the items to be sorted are delivered for storage; the tag information read by the second radio frequency tag reader **190** is used for feeding, that is, to determine the feeding tables to which the items to be sorted that are to be moved for classified storage.

[0066] In an optional implementation of this embodiment, this implementation may be combined with the optional solutions in one or more of the foregoing embodiments. FIG. 4 is a schematic diagram of another sorting system provided in an embodiment of the present disclosure. Referring to FIG. 4, the sorting system **100** further includes at least one handling robot **1110** or item conveying facility **1120**.

[0067] The control server **130** sends a feeding instruction to at least one handling robot **1110** or item conveying facility **1120** according to the feeding table **160** corresponding to the items to be sorted that are to be moved to the feeding table.

[0068] The handling robot **1110** receives the feeding instruction, and in response to the feeding instruction, moves the items to be sorted that are to be moved to the feeding table, to the corresponding part feeding table **160**; or, [0069] the item conveying facility **1120** receives the feeding instruction, and in response to the feeding instruction, moves the items to be sorted that are

to be moved to the feeding table, to the corresponding feeding table **160**.

[0070] In this implementation, a plurality of handling robots **1110** are deployed between the unloading area where the items to be sorted are stacked (for example, the position of the item buffer facility **1100**) and the feeding table **160**; or, one or more item conveying facilities **1120** are deployed; or, a combination of handling robots **1110** and item conveying facilities **1120** is deployed. After generating the feeding instruction, the control server **130** sends the feeding instruction to at least one of the handling robots **1110** and the item conveying facilities **1120**.

[0071] In this implementation, in an optional example, referring to FIG. 4, for the handling robot **1110**, the handling robot **1110** can transport the items to be sorted buffered in the item buffer facility **1100** to the corresponding feeding table according to the feeding instruction **160** for temporary storage. In another optional example, referring to FIG. 4, for the item conveying facility **1120**, the item conveying facility **1120** can receive the feeding instruction sent by the control server **130**, and convey the items to be sorted buffered in the item buffer facility **1100** to the corresponding feeding table according to the feeding instruction **160** for temporary storage. FIG. 5 is a schematic diagram showing a part of an item conveying facility in a sorting system provided in an embodiment of the present disclosure. Referring to FIG. 5, the item conveying facility **1120** includes an entrance, a plurality of exits, and a conveying part connecting the entrance and the multiple exits. The item to be sorted can enter the item conveying facility **1120** through the entrance, and is conveyed by the conveying part and comes out of the item conveying facility **1120** from one of the multiple exits. There is a preset correspondence relationship between the multiple exits and the multiple feeding tables **160**. The item conveying facility **1120** guides the item to be sorted to a target exit corresponding to the corresponding feeding table **160** according to the feeding instruction, so as to convey the item to be sorted to the corresponding feeding table **160**.

[0072] In an optional implementation of this embodiment, this embodiment may be combined with the optional solutions in one or more of the foregoing embodiments. Referring to FIG. 4, the sorting system further includes a second grabbing component **1130**. The second grabbing component **1130** is configured to grab the item to be sorted that is located in the unloading area and to be moved to the feeding table, and place the grabbed item to be sorted on the handling robot **1110** or the item conveying facility **1120**, so that the handling robot **1110** or the item conveying facility **1120** supplies the item to be sorted to the corresponding feeding table **160**. In one embodiment, the second grabbing component **1130** may include a robotic arm, and a robotic gripper or a suction cup is provided on the robotic arm, and the items to be sorted may be grabbed by the robotic gripper or the suction cup.

[0073] In this implementation, in an optional example, referring to FIG. 4, the second radio frequency tag reader **190** is disposed on the second grabbing component **1130**. When the second grabbing component **1130** grabs the item to be sorted that is to be moved to the feeding table from the item buffer facility **1100**, the second radio frequency tag reader **190** directly reads tag information in the radio frequency tag of the grabbed item to be sorted and send the read tag information in the radio frequency tag of the item to be sorted to the control server **130**. In another optional example, referring to FIG. 4, the second radio frequency tag reader **190** is disposed on the handling robot **1110** or the item conveying facility **1120**. When the second grabbing component **1130** grabs the item to be sorted that is to be moved to the feeding table from the item buffer facility **1100** and places the grabbed item to be sorted on the handling robot **1110** or the item conveying facility **1120**, the second radio frequency tag reader **190** located on the handling robot **1110** or the item conveying facility **1120** directly reads the tag information in the radio frequency tag of the grabbed item to be sorted, and sends the read tag information in the radio frequency tag of the item to be sorted to the control server **130**.

[0074] In this embodiment, the setting position of the second radio frequency tag reader shown in FIG. 3 may deviate, and the setting positions of the second radio frequency tag reader and the second grabbing component shown in FIG. 4 may deviate from the positions described in text. For

specific settings, please refer to the text descriptions for FIG. 3 and FIG. 4.

[0075] According to the sorting solution provided in embodiments of the present disclosure, the tag information in the radio frequency tags of the items to be sorted that are to be moved to the feeding table is automatically read by the second radio frequency tag reader, and the control server automatically instructs supplying the items to be sorted to the matching feeding table for buffering according to the tag information read by the second radio frequency tag reader. With the solution of this embodiment, since the radio frequency reader has no requirement on the reading direction, no matter in what posture the items to be sorted are placed, the second radio frequency tag reader can read the tag information in the radio frequency tags of the items to be sorted without manual operation, so as to ensure that the tag information are automatically recognized and sent when the items to be sorted are grabbed. In this way, the control server can determine the sorting destination of the grabbed item to be sorted in advance, and automatically supply the item to be sorted to the corresponding feeding table, so as to realize the unmanned operation of the feeding process, which saves the time wasted by manually aligning the barcodes of commodities or packages in the feeding link to identify the barcodes of the commodities or packages, improves the feeding efficiency and reduces the labor cost in the feeding link.

[0076] FIG. 6 is a schematic diagram showing a part of a floor-type sorting scenario including a collection table provided in an embodiment of the present disclosure. Referring to FIG. 6, in the traditional way, the collection table requires a staff to check the items to be sorted in the collection container. On the basis of the foregoing embodiments, this implementation may be combined with optional solutions in one or more of the foregoing embodiments. FIG. 7 is a schematic diagram of another sorting system provided in an embodiment of the present disclosure. Referring to FIG. 7, the sorting system **100** further includes at least one third radio frequency tag reader **1140** and at least one collection table **1150**. The third radio frequency tag reader **1140** is disposed at a preset position of the collection table **1150**.

[0077] The third radio frequency tag reader **1140** is configured so that when the collection container **170** containing the items to be sorted is close to the third radio frequency tag reader **1140**, the third radio frequency tag reader **1140** reads the tag information of the radio frequency tags of all the items to be sorted in the collection container **170**, and send the read tag information of all the radio frequency tags to the control server **130**.

[0078] The control server **130** is further configured to determine whether all the items to be sorted **150** meet a sorting condition according to the tag information of all the radio frequency tags, and instruct handling the collection container **170** containing the items to be sorted to the collection table **1150** when determining that the items to be sorted **150** meet the sorting condition.

[0079] In this embodiment, referring to FIG. 7, the third radio frequency tag reader **1140** is disposed at a preset position between the collection table **1150** and the plurality of collection containers **170** to ensure that the collection container **170** containing the items to be sorted will first pass the third radio frequency tag reader **1140** and then reach the collection table **1150**. In this way, when the collection container **170** containing the items to be sorted passes the third radio frequency tag reader **1140**, the third radio frequency tag reader **1140** reads the tag information of all the radio frequency tags in the collection container **170**. Optionally, the third radio frequency tag reader **1140** is an RFID identification gate. When the collection container **170** containing the items to be sorted passes through the RFID identification gate, the RFID identification gate will automatically read the tag information of all the radio frequency tags in the collection container **170**, and send the tag information of all radio frequency tags to the control server **130**.

[0080] In this embodiment, there may be deviations in the setting position of the third radio frequency tag reader **1140** shown in FIG. 7. For the setting situation of the third radio frequency tag reader **1140**, please refer to the text description for FIG. 7.

[0081] In this embodiment, the control server **130** determines whether the sorting destination of each item to be sorted contained in the collection container **170** matches the sorting destination of

the current collection container according to the tag information of all the radio frequency tags, and determines whether the number of items to be sorted in the collection container **170** is correct, and determines whether the sorting condition is met in combination with the above aspects. If it is determined that the items to be sorted meet the sorting condition, the sorting robot is instructed to handle the collection container containing the items to be sorted to the collection table **1150**, so as to be packed and loaded, that is, the sorting can be completed. If it is determined that the items to be sorted do not meet the sorting condition, it indicates that multiple items to be sorted in the collection container **170** do not meet the sorting requirements, and at this time, the collection container **170** containing the items to be sorted is transferred and exception handling is performed. By adopting the above method, there is no need to manually review the items to be sorted in the collection container **170** at the collection table **1150**, and the review work can be automatically completed when the collection container **170** reaches the collection table **1150**, thereby speeding up the collection efficiency, further improving the sorting efficiency of the entire sorting process, and reducing the labor cost of the collection link, thereby reducing the labor cost of the entire sorting process.

[0082] FIG. **8** is a flowchart of a sorting method provided in an embodiment of the present disclosure. The technical solution of the present embodiment can be applied to the unmanned sorting of commodities or packages, and the method can be applied to an unmanned sorting system provided in any embodiment of the present disclosure. As shown in FIG. **7**, the sorting method in the embodiment of the present disclosure includes the following steps.

[0083] At **S810**, the first radio frequency tag reader reads the tag information of each radio frequency tag in at least one radio frequency tag, and sends the read tag information to the control server. The at least one radio frequency tag is respectively configured on at least one item to be sorted placed on at least one feeding table, and the tag information of each radio frequency tag includes sorting destination information of the item to be sorted where each radio frequency tag is located.

[0084] Based on the technical solutions of the above embodiments, optionally, the first radio frequency tag reader is disposed at a preset position of the feeding table; or, the first radio frequency tag reader is disposed at a preset position of the sorting robot; or, the first radio frequency tag reader is disposed between the feeding table and the at least one container, and the distance between the first radio frequency tag reader and the preset position of the feeding table is less than the preset value.

[0085] On the basis of the technical solutions of the foregoing embodiments, optionally, the method further includes followings.

[0086] The first grabbing component grabs the items to be sorted from the feeding table, and places the grabbed items to be sorted on the at least one sorting robot, wherein, the first grabbing component is disposed on the feeding table. In the case where the first radio frequency tag reader is disposed at a preset position of the feeding table, the first radio frequency tag reader is disposed on the first grabbing component.

[0087] Based on the technical solutions of the foregoing embodiments, optionally, the at least one sorting robot includes a plurality of delivery mechanisms, and the delivery mechanisms are configured to deliver at least one item to be sorted on the sorting robot.

[0088] At **S820**, the control server determines the sorting destination of the item to be sorted where each radio frequency tag is located according to the tag information read by the first radio frequency tag reader, and determine a target collection container from at least one collection container according to the sorting destination, and sends a delivery instruction to at least one sorting robot.

[0089] At **S830**, in response to the delivery instruction, the at least one sorting robot delivers each item to be sorted where each radio frequency tag is located, obtained from the at least one feeding table, to a corresponding target collection container.

[0090] Based on the technical solutions of the above embodiments, optionally, in response to the delivery instruction, the at least one sorting robot delivers at least one item to be sorted where each radio frequency tag is located, obtained from the at least one feeding table, to the target collection container as follows. The at least one sorting robot, in response to the delivery instruction, delivers the items to be sorted where each radio frequency tag is located, obtained from the at least one feeding table, to the target collection container through the delivery mechanisms of the at least one sorting robot.

[0091] On the basis of the technical solutions of the above embodiments, optionally, each collection container corresponds to at least one sorting destination, and collection containers corresponding to the same sorting destination are arranged adjacent to each other; the at least one feeding table corresponds to at least one sorting destination.

[0092] On the basis of the technical solutions of the foregoing embodiments, optionally, the sorting method further includes: [0093] reading, by the second radio frequency tag reader, the tag information in the radio frequency tags of the items to be sorted that are to be moved to the feeding table, and sending the read tag information to the control server; [0094] determining, by the control server, the sorting destination of the item to be sorted that is to be moved to the feeding table according to the tag information read by the second radio frequency tag reader, and determining the feeding table corresponding to the item to be sorted that is to be moved to the feeding table, from the at least one feeding table according to the sorting destination, so that the item to be sorted that is to be moved to the feeding table is moved to the corresponding feeding table.

[0095] On the basis of the technical solutions of the foregoing embodiments, optionally, the sorting method further includes: [0096] sending, by the control server, a feeding instruction to at least one handling robot or at least one item conveying facility according to the corresponding feeding table of the item to be sorted that is to be moved to the feeding table; [0097] receiving, by the handling robot, the feeding instruction, and moving the item to be sorted that is to be moved to the feeding table, to the corresponding feeding table in response to the feeding instruction; or, [0098] receiving, by the item conveying facility, the feeding instruction, and moving the item to be sorted that is to be moved to the feeding table, to the corresponding feeding table in response to the feeding instruction.

[0099] On the basis of the technical solutions of the foregoing embodiments, optionally, the sorting method further includes: [0100] grabbing, by the second grabbing component, the item to be sorted that is to be moved to the feeding table, and placing the grabbed item to be sorted on the handling robot or the item conveying facility.

[0101] On the basis of the technical solutions of the above embodiments, optionally, the second radio frequency tag reader is disposed on the second grabbing component.

[0102] On the basis of the technical solutions of the foregoing embodiments, optionally, the second radio frequency tag reader is disposed on the handling robot or the item conveying facility.

[0103] On the basis of the technical solutions of the above-mentioned embodiments, optionally, the sorting method further includes: receiving, by the item buffer facility, the items to be sorted that are transported by the transport equipment and are to be moved to the feeding table, and buffering the items to be sorted that are to be moved to feeding table, wherein the second radio frequency tag reader is disposed at a preset position between the item buffer facility and the feeding table.

[0104] On the basis of the technical solutions of the foregoing embodiments, optionally, the sorting method further includes: [0105] in the case where the distance between the collection container containing the items to be sorted and the third radio frequency tag reader is less than the preset distance threshold, reading by the third radio frequency tag reader, the tag information of all radio frequency tags in the collection container, and sending the tag information of all the radio frequency tags to the control server; wherein, the third radio frequency tag reader is disposed at the preset position of the collection table; [0106] determining, by the control server, whether all the items to be sorted meet the sorting condition according to the tag information of all the radio

frequency tags; and, in the case where it is determined that all the items to be sorted meet the sorting condition, instructing the handling robot to handle the collection container containing the items to be sorted to collection table.

[0107] Based on the technical solutions of the foregoing embodiments, optionally, the sorting condition include at least one of the following: whether the sorting destination of the items to be sorted contained in the collection container matches the sorting destination of the collection container; and whether the number of items to be sorted contained in the collection container is correct.

[0108] The sorting method provided in embodiments of the present disclosure can be applied to the sorting system provided in any of the above embodiments of the present disclosure, and has the corresponding functions of the sorting system. For the technical details not described in detail in the above embodiments, please refer to the sorting system provided in any of the embodiments of this disclosure.

Claims

1. A sorting system, comprising: a sorting platform, a first tag reader, a control server, at least one sorting robot, at least one feeding table and a plurality of collection containers, wherein the sorting platform comprises one or more layers, and the at least one sorting robot runs on the sorting platform; each feeding table is configured to place at least one item to be sorted; and each item to be sorted is provided with a tag comprising sorting destination information of the item to be sorted; the first tag reader is configured to read tag information in the tag, and send the tag information to the control server; the plurality of collection containers are divided into at least one collection container group; the control server is configured to determine a sorting destination of the item to be sorted according to the tag information comprising the sorting destination information, determine a target collection container corresponding to the item to be sorted from the collection container group according to the sorting destination, and send a delivery instruction and a delivery path of the item to be sorted planed according to a position of the target collection container to the sorting robot; and the sorting robot is configured to, in response to the delivery instruction received from the control server, deliver the item to be sorted to the target collection container according to the delivery path.
2. The system of claim 1, wherein one collection container group is associated with one feeding table, and a plurality of sorting destinations corresponding to the feeding table are the same as sorting destinations corresponding to a plurality of collection containers in the collection container group.
3. The system of claim 1, wherein one collection container group is associated with a plurality of feeding tables, the plurality of feeding tables correspond to a plurality of sorting destinations, and items to be sorted on the plurality of feeding tables are delivered to a plurality of collection containers in the collection container group.
4. The system of claim 1, further comprising: an item buffer facility, wherein the item buffer facility is configured to receive an item to be sorted that is transported by a transport equipment and is to be moved to a feeding table, and buffer the item to be sorted that is to be moved to the feeding table.
5. The system of claim 1, further comprising: an unloading area, wherein the at least one item to be sorted is transported to the unloading area by using a transport equipment, and unloaded in the unloading area, and the item to be sorted in the unloading area is an item to be sorted that is to be moved to the feeding table.
6. The system of claim 1, further comprising a first grabbing component disposed on the feeding table, the first grabbing component being configured to receive the at least one item to be sorted, and to place the at least one item to be sorted on the at least one sorting robot.

7. The system of claim 1, further comprising: at least one item conveying facility, wherein the control server is further configured to send a feeding instruction to the at least one item conveying facility according to a feeding table corresponding to an item to be sorted that is to be moved to the feeding table; and the item conveying facility is configured to receive the feeding instruction, and in response to the feeding instruction, move the item to be sorted that is to be moved to the feeding table to the feeding table.

8. The system of claim 1, further comprising at least one handling robot, wherein the control server is further configured to send a feeding instruction to the at least one handling robot according to a feeding table corresponding to an item to be sorted that is to be moved to the feeding table; and the handling robot is configured to receive the feeding instruction, and in response to the feeding instruction, move the item to be sorted that is to be moved to the feeding table to the feeding table.

9. The system of claim 4, further comprising: a second tag reader; wherein the second tag reader is configured to read the tag information in the tag of the item to be sorted that is to be moved to the feeding table, and send the read tag information to the control server, in which the second tag reader is disposed at a preset position between the item buffer facility and the feeding table; and the control server is further configured to determine the feeding table corresponding to the item to be sorted that is to be moved to the feeding table according to the tag information.

10. A sorting system, comprising: a first tag reader, a control server, at least one sorting robot, at least one feeding table, a plurality of collection container groups, a first grabbing component and a sorting platform, wherein each feeding table is configured to place at least one item to be sorted; and each item to be sorted is provided with a tag comprising sorting destination information of the item to be sorted; the first tag reader is configured to read tag information in the tag, and send the tag information to the control server to determine a sorting destination of the item to be sorted; the first grabbing component is configured to receive the at least one item to be sorted, and to place the at least one item to be sorted on the at least one sorting robot; and the at least one sorting robot is configured to, in response to a delivery instruction received from the control server, deliver the at least one item to be sorted to a target collection container in a collection container group according to a delivery path.

11. The system of claim 10, wherein the control server is configured to determine the sorting destination of the item to be sorted according to the tag information comprising the sorting destination information, determine the target collection container corresponding to the item to be sorted from the collection container group according to the sorting destination, and send the delivery instruction and the delivery path of the item to be sorted planned according to a position of the target collection container to the sorting robot, wherein the first tag reader is provided in a preset position of the feeding table.

12. The system of claim 10, further comprising: an unloading area, wherein the at least one item to be sorted is transported to the unloading area by using a transport equipment, and unloaded in the unloading area, and the item to be sorted in the unloading area is an item to be sorted that is to be moved to the feeding table.

13. The system of claim 10, further comprising: at least one item conveying facility, wherein the control server is further configured to send a feeding instruction to the at least one item conveying facility according to a feeding table corresponding to the item to be sorted that is to be moved to the feeding table; and the item conveying facility is configured to receive the feeding instruction, and in response to the feeding instruction, move the item to be sorted that is to be moved to the feeding table to the feeding table.

14. A sorting system, comprising: a first tag reader, a control server, at least one sorting robot, at least one feeding table, and a plurality of collection containers, wherein each feeding table is configured to place at least one item to be sorted; and each item to be sorted is provided with a tag comprising sorting destination information of the item to be sorted; the first tag reader is configured to read tag information in the tag, and send the tag information to the control server; the

control server is configured to determine a sorting destination of the item to be sorted according to the tag information comprising the sorting destination information, determine a target collection container corresponding to the item to be sorted according to the sorting destination, and send a delivery instruction and a delivery path of the item to be sorted planned according to a position of the target collection container to the sorting robot; and the sorting robot is configured to, in response to the delivery instruction received from the control server, deliver the item to be sorted to the target collection container according to the delivery path.

15. The system of claim 14, wherein the plurality of collection containers are divided into at least one collection container group.

16. The system of claim 14, further comprising: a first grabbing component and at least one of sorting platform, wherein the sorting robot runs on the sorting platform, and the first grabbing component is configured to receive the at least one item to be sorted, and place the at least one item to be sorted on the at least one sorting robot.

17. The system of claim 16, wherein each grid opening of the sorting platform corresponds to the collection container, horizontal and vertical gaps between adjacent collection containers correspond to horizontal gaps and vertical gaps between adjacent grid openings on the sorting platform, and the horizontal gaps and vertical gaps between the adjacent grid openings on the sorting platform constitute a walking passage of the sorting robot on the sorting platform.

18. The system of claim 14, further comprising: at least one item conveying facility, wherein the control server is further configured to send a feeding instruction to the at least one item conveying facility according to a feeding table corresponding to an item to be sorted that is to be moved to the feeding table; and the item conveying facility is configured to receive the feeding instruction, and in response to the feeding instruction, move the item to be sorted that is to be moved to the feeding table to the feeding table.

19. The system of claim 18, wherein the item conveying facility comprises an entrance, a plurality of exits, and a conveying part connecting the entrance and the plurality of exits, the item to be sorted that is to be moved to the feeding table enters the item conveying facility through the entrance, conveyed by the conveying part and comes out of the item conveying facility from one of the plurality of exits, wherein there is a preset correspondence relationship between the plurality of exits and a plurality of feeding tables.

20. The system of claim 14, wherein the at least one sorting robot comprises a plurality of delivery mechanisms, and the delivery mechanism is configured to deliver the at least one item to be sorted on the sorting robot, wherein in the case that the sorting robot carries a plurality of items to be sorted and the plurality of items to be sorted have a same sorting destination, the sorting robot delivers the plurality of items to be sorted to a same collection container according to the sorting destination; and in the case that the sorting robot carries the plurality of items to be sorted and the plurality of items to be sorted have different sorting destinations, the sorting robot delivers the plurality of items to be sorted to different collection containers in turn according to the different sorting destinations.
